

# Data Bible: Bolt Datathon

## 1. Overview

Brief description of the dataset and its purpose.

- **Project:** Vancouver City FC Revenue Insight
- **Source:** Bolt Case Package
- **Date Range:** Dec. 2023 - Dec. 2024

## 2. Data Sources

| Source Name              | Description                                                 | Format |
|--------------------------|-------------------------------------------------------------|--------|
| fanbase_clean.csv        | Member profiles and demographics                            | CSV    |
| cleaned_merch.csv        | Purchase history and item categories                        | CSV    |
| clean_stadium.csv        | Stadium operations incl. revenue sources                    | CSV    |
| merch_fanbase_merged.csv | Merged file of merch and fanbase info (Outer, on Member ID) | CSV    |

## 3. Data Dictionary

Fanbase Engagement ( fanbase\_clean.csv ) 70000 records

| Column Name        | Type                 | Description                                  |
|--------------------|----------------------|----------------------------------------------|
| Member_ID          | integer              | Unique membership identifier (ID)            |
| Customer_Age_Group | string (Ordinal)     | Age bracket                                  |
| Customer_Region    | string (Categorical) | Country or broader region the member is from |
| Seasonal_Pass      | boolean              | Whether customer holds a seasonal pass       |

Customer\_Age\_Group

Value

'<18'

'18-25'

|  | Value   |
|--|---------|
|  | '26-40' |
|  | '41-60' |
|  | '60+'   |

Customer\_Region

| Value         |
|---------------|
| 'Canada'      |
| 'USA'         |
| 'Japan'       |
| 'India'       |
| 'China'       |
| 'South Korea' |
| 'UK'          |

Stadium Operations( clean\_stadium.csv ) 144 records

| Column Name | Type                 | Description                                                   |
|-------------|----------------------|---------------------------------------------------------------|
| Month       | integer (Ordinal)    | The month within the 2024 Calendar Year                       |
| Revenue     | integer              | Revenue amount generated by assigned source during that month |
| Source      | string (Categorical) | Specific source of revenue or cost                            |

Month

| Code | Value    |
|------|----------|
| 1    | January  |
| 2    | February |
| 3    | March    |
| 4    | April    |
| 5    | May      |
| 6    | June     |
| 7    | July     |
| 8    | August   |

| Code | Value     |
|------|-----------|
| 9    | September |
| 10   | October   |
| 11   | November  |
| 12   | December  |

## Source

| Value        |
|--------------|
| 'Upper Bowl' |
| 'Lower Bowl' |
| 'Season'     |
| 'Premium'    |
| 'Food'       |
| 'Staff'      |
| 'Concert'    |

## Merchandise Purchase ( cleaned\_merch.csv ) 67877 records

| Column Name        | Type                  | Description                                                                                                                                                                                             |
|--------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product_ID         | Integer (Categorical) | Unique product identifier                                                                                                                                                                               |
| Barcode            | Integer               | The barcode (e.g., UPC/EAN) associated with the exact SKU sold to an individual customer; this is the number scanned at checkout.                                                                       |
| Item_Category      | Categorical           | Type of merchandise                                                                                                                                                                                     |
| Item_Name          | Categorical           | Specific item name                                                                                                                                                                                      |
| Size               | Categorical           | Item size                                                                                                                                                                                               |
| Unit_Price         | Integer (Ordinal)     | Price per unit (CAD)                                                                                                                                                                                    |
| Customer_Age_Group | Categorical           | Estimated age bracket of customer, may differ from purchaser age                                                                                                                                        |
| Customer_Region    | Categorical           | Domestic vs international                                                                                                                                                                               |
| Promotion          | Boolean               | Indicates whether a specific promotional campaign led to the purchase. If the customer engaged with a tracked promotion (e.g., clicked an ad) and then purchased, this is TRUE; otherwise, it is FALSE. |

| Column Name  | Type        | Description                                                                                                                             |
|--------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Channel      | Categorical | Sales channel                                                                                                                           |
| Selling_Date | Date        | Date and time of transaction                                                                                                            |
| Arrival_Date | Date        | For online orders, the date and time the product was delivered to the customer. This field is left blank (or NA) for in-store purchases |

#### Item\_Category

Cap

Hoodie

Jersey

Mug

Poster

Scarf

Youth Hoodie

Youth Jersey

#### Product\_ID

#### Item\_Name

10000001 Classic Cap

10000002 Retro Cap

10000003 Training Hoodie

10000004 Home Jersey

10000005 Away Jersey

10000006 Retro Jersey

10000007 Club Mug

10000008 Logo Mug

10000009 Player Poster

10000010 Team Poster

10000011 Home Scarf

10000012 Away Scarf

10000013 Youth Team Hoodie

10000014 Youth Training Hoodie

10000015 Youth Away Jersey

10000016 Youth Retro Jersey

#### Size

L

M

S

XL

XXL

YL

YS

YM

NaN

#### Customer\_Age\_Group

<18

18–25

26–40

41–60

60+

#### Channel

Online

Team Store

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## 4. Cleaning & Transformation Logic

- outer merged merch and fanbase on shared member IDs
  - imputed ages from fanbase data into merch for matching member IDs (ages are estimated in merch, but not in fanbase)
  - regions were standardized by the following priorities:
    1. if item was purchased through a team store, the member region was set to Domestic
    2. if member region is consistent between duplicates, leave it be
    3. if member region is inconsistent between duplicates, classify as Domestic or International based on fanbase information Note: fanbase information was not used to begin with, as some of these values were inconsistent with channel of purchase
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